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Quantifying Conversational Reliability of Large Language Models under Multi-Turn Interaction

Jiyoon Myung
Samsung SDS
jiyoon0424@gmail.com

Abstract

Large Language Models (LLMs) are increasingly deployed in real-world applications where users engage in extended, mixed-topic conversations that depend on prior context. Yet, their reliability under realistic multi-turn interactions remains poorly understood. We conduct a systematic evaluation of conversational reliability through three representative tasks that reflect practical interaction challenges: (1) maintaining global constraints across topic shifts, (2) selecting the correct tool or agent amid interleaved intents, and (3) tracking structured entities under revisions and distractions. Each task pairs single-turn and multi-turn settings, allowing us to quantify reliability degradation under extended dialogue. Across both commercial and open-source models, we observe substantial declines in reliability, particularly for smaller models. Error analyses reveal recurring failure modes such as instruction drift, intent confusion, and contextual overwriting, which compromise dependable behavior in operational systems. Our findings highlight the need for stress-testing LLMs for conversational reliability and developing more robust evaluation methods for trustworthy deployment.

Introduction

Deployed conversational systems must operate reliably in messy, multi-turn environments: users shift topics, interleave irrelevant content, and revise their goals mid-dialogue. Models are therefore expected to remain consistent and robust as conversations grow longer and more context-dependent. Failures in these basic interactive abilities can seriously undermine the usability of a system.

Empirical studies show that large language models (LLMs) often struggle under such conditions. Multi-turn analyses reveal substantial degradation in reliability compared to single-turn prompts (Laban et al. 2025), while long-context evaluations expose weaknesses such as the "lost in the middle" effect (Liu et al. 2023). Several benchmarks—such as Multi-IF (He et al. 2024), StructFlowBench (Li et al. 2025), MMMT-IF (Epstein et al. 2024), and MINT (Wang et al. 2024)—probe aspects of conversational robustness, including instruction consistency and tool use, but they often focus on abstract challenges or rely on subjective judgments. This leaves open the question of how to objectively evaluate concrete behaviors required in practice.

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We address this gap by introducing three compact, pass/-fail evaluable tasks that reflect core requirements for real-world assistants:

- **Instruction Following under extended conversation**: enforcing a global style constraint despite distracting turns.
- **Tool Selection in mixed-topic dialogues**: routing each request to the correct tool when multiple intents are interleaved.
- **Entity Extraction under revisions and distractions**: tracking the user's final structured intent despite changes of mind, chit-chat, or irrelevant mentions.

Each task is paired with a single-turn counterpart, allowing us to isolate and quantify the degree of *reliability degradation* that occurs under extended, mixed-topic interactions. This paired design enables reproducible and objective assessment of how conversational reliability deteriorates across model families and dialogue lengths. By comparing large commercial LLMs with smaller open-source counterparts, we further reveal capacity-dependent vulnerabilities that have direct implications for real-world deployment and safety.

In summary, our study bridges the gap between research benchmarks and practical evaluation by providing deterministic, reproducible tests of conversational robustness—highlighting where current models fail to sustain reliable behavior over time.

Experiments

Our goal is to quantify how performance degrades when tasks are embedded in extended, mixed-topic conversations, compared to their single-turn counterparts. We therefore define three representative tasks motivated by real-world service scenarios, construct synthetic single-turn and multi-turn dialogues, and evaluate a range of models from commercial LLMs to smaller open-source SLMs. We report task accuracy and analyze common error patterns, highlighting the operational risks that arise in multi-turn interactions.

Task Scenarios

We focus on three multi-turn challenges that frequently arise in real-world conversational systems:

Figure 1: Single-turn vs Multi-turn accuracy across three evaluation tasks. Each panel shows a task example on the left and model accuracy on the right. Performance drops most severely in **Instruction Following**, while **Entity Extraction** remains relatively robust, and **Tool Selection** shows mixed degradation depending on model size.

Instruction Following. A length constraint is specified at the beginning of the dialogue (e.g., "always answer in at most 5 sentences"). The conversation then continues with unrelated topics over multiple turns, and the final user request is deliberately phrased to elicit a long and detailed answer. The evaluation measures whether the model consistently respects the global style constraint throughout the dialogue, even when pressured to violate it. This represents scenarios where chatbots must reliably enforce prescribed formatting rules, such as staying concise or avoiding certain tokens.

Tool Selection. At specific turns, the model must select the correct tool from a fixed set: `[Weather, News, Calculator, Stock, Recipe, Dictionary]`. In the single-turn setting, the user query directly corresponds to a single tool. In the multi-turn setting, conversations are explicitly constructed to *mix multiple topics* from the tool list, so that the model must correctly route each request in a more challenging environment. This reflects real-world situations such as intent classification in digital assistants or routing queries to the right component in multi-agent systems.

Entity Extraction. The task is to extract the final structured slots in a restaurant reservation scenario: `(date, time, number of people)`. In single-turn cases, the reservation request is stated directly. In multi-turn cases, we introduce realistic complications: users may change their mind multiple times (*change in mind*), engage in unrelated small talk (*intermediate chit chat*), or mention other people's reservations (*multiple mention*). The model must correctly track the user's final intent and output the canonical reservation values. This setting mirrors practical needs such as extracting parameters for executing tools (e.g., calendar or booking APIs) or supporting real agents that must handle evolving goals in natural conversations.

Data Generation

For each task, we generated paired single-turn and multi-turn dialogue datasets.

- **Dialogue generation**: We used GPT-5 to synthesize dialogues, controlling for dialogue length, number of topic shifts, and frequency of modifications. Synthetic generation allowed us to systematically vary factors such as distraction density or revision frequency while keeping overall style consistent.
- **Single-turn vs. multi-turn**: Each instance was created in two variants—one where the relevant request was posed directly (single-turn), and one where it was embedded within a longer conversation with distractions, topic changes, or corrections (multi-turn).
- **Task-specific constraints**: For *Instruction Following*, dialogues contained between 5 and 15 turns. This range was chosen to roughly approximate short-to-moderate real customer service sessions, where most exchanges end within a dozen turns. For *Tool Selection*, dialogues ranged from 6 to 16 turns, and the number of distinct relevant tools was randomized between 2 and 6 to mimic real assistants that must juggle several intents in a single session. For *Entity Extraction*, each dialogue randomly combined conditions such as *change in mind*, *intermediate chit chat*, and *multiple mention*, reflecting diverse patterns observed in reservation-style conversations.

- **Size**: For each task, we generated approximately 100 dialogues per condition, yielding about 600 evaluation cases in total across all tasks.
- **Annotation**: Ground-truth labels were automatically derived during generation (e.g., correct tool, final reservation details) and verified by human inspection on a sample basis.
- **Availability**: The generated datasets will be released on HuggingFace after publication to support reproducibility and further research.

Models Evaluated

We evaluated a diverse set of language models covering both commercial and open-source deployments. All models were accessed via their respective official APIs, and we fixed the decoding temperature to 0 to ensure deterministic outputs.

- **Commercial LLMs**: GPT-4o, GPT-4o-mini, Gemini-2.5-Flash.
- **Open-source SLMs**: Qwen-8B, Qwen-32B, Ministral-8B, Mistral-small-24B, Gemma-3-12B.

Evaluation Metrics

As all tasks were designed with clear pass/fail criteria, we adopt *accuracy* as the primary metric to ensure clarity, replicability, and ease of interpretation across diverse models. Unlike open-ended generation or preference-based evaluation, our tasks do not benefit from graded or subjective metrics.

- **Instruction Following**: An output is correct if it satisfies the constraint of containing at most five sentences. Any response exceeding this length limit is considered incorrect.
- **Tool Selection**: An output is correct if the tool selected by the model matches the ground-truth tool for that turn.
- **Entity Extraction**: An output is correct if the extracted `(date, time, number of people)` exactly matches the ground-truth values provided with the dialogue.

Results

Overall Results

Table 1 summarizes the main experimental results. Across all tasks, we observe a consistent degradation when moving from single-turn to multi-turn settings. McNemar tests confirm that these performance gaps are statistically significant across all three tasks (see Appendix for full results). This confirms that multi-turn dialogue introduces substantial additional difficulty across model scales.

Instruction following exhibited the largest degradation overall. While most models performed strongly in the single-turn setting, accuracy dropped sharply once the task was embedded in multi-turn dialogues. This decline was evident even for commercial LLMs (e.g., GPT-4o falling from 96% to 63%, Gemini-2.5-Flash from 96% to 89%), and was even more severe for smaller models (e.g., GPT-4o-mini at 24%, Qwen3-8B at 27%). These results suggest that maintaining global constraints over extended conversations remains a fundamental challenge, regardless of model capacity.

Tool selection, by contrast, showed strong robustness among commercial models. Systems such as GPT-4o and Gemini-2.5-Flash maintained very high accuracy ($\geq$97%) even in multi-turn settings. However, smaller open-source models displayed substantial degradation, particularly when multiple tools were relevant in the same dialogue. For example, Qwen3-32B dropped to 47% and Ministral-8B to 37%. This indicates that explicit grounding to a fixed tool set is relatively easy for larger models, but smaller ones struggle to maintain consistency under higher conversational complexity.

Entity extraction emerged as the most robust task across models. In single-turn reservations, nearly all models achieved near-perfect accuracy (96–100%). Even in multi-turn dialogues, where user changes of mind and distractions were introduced, performance remained relatively high (typically 84–89%, with the lowest at 79%). This resilience likely stems from the structured nature of the target fields—`(date, time, number of people)`—which are expressed as explicit numbers or short phrases. As a result, models could reliably capture the final slot values with limited ambiguity. We hypothesize that if the extraction targets had required richer contextual understanding, for example, mapping free-form user mentions like "the pizza with pineapple" to a canonical menu item (*Hawaiian pizza*), the degradation would have been much more pronounced.

We also note model-specific patterns. **Mistral** was particularly weak at the instruction-following task, even in single-turn scenarios (27-58%), suggesting difficulty in adhering to length constraints. On the other hand, **Gemma** models demonstrated remarkable robustness in tool selection, with multi-turn performance staying almost at the same level as single-turn. Interestingly, larger models (e.g., GPT-4o, Gemma-3-27B, Qwen-32B) showed smaller performance gaps between single-turn and multi-turn than smaller ones (e.g., GPT-4o-mini, Mistral-8B), underscoring the role of model capacity in sustaining performance under long-horizon interactions.

Overall, these findings reinforce that while multi-turn dialogue universally degrades accuracy, the degree of impact depends strongly on the task type and model family, with global instruction maintenance emerging as the most challenging dimension.

Detailed Error Analysis

To understand the causes of reliability degradation, we analyze results by conversation length, task complexity, and entity extraction scenario type (see Appendix for full tables).

Conversation length alone is not a dominant factor for instruction-following degradation. Accuracy fluctuates but does not consistently worsen in longer dialogues; at 10 turns it even peaks at 96%. This suggests that failures arise not merely from context length but from specific distractors or competing user demands.